Design Document (DD)

Energy-Efficient Dual Hydroponic System for Organic/Inorganic Capsicum Chili Farming

Project ID: 25-26J-035

Individual component: Design and Implement an Automated Organic NFT Hydroponic System

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1. Introduction

1.1 Purpose

The purpose of this checklist is to verify and demonstrate the current progress of the project at Progress Presentation 1 stage. At this milestone, the focus is on system setup, sensor integration, data acquisition, cloud connectivity, and basic monitoring, while automation and nutrient control are planned for later stages.

1.2 Scope

This document covers:

- NFT hydroponic structure implementation
- Sensor integration and real-time data acquisition
- ESP32 firmware for reading sensors
- Firebase cloud data upload
- Mobile/dashboard visualization of sensor readings

This document **does not yet include**:

- Fuzzy logic automation
- Compost maturity classification logic
- Automated nutrient dosing decisions

These are planned for subsequent milestones.

1.3 Definitions, Acronyms, and Abbreviations

Term	Meaning
NFT	Nutrient Film Technique
IoT	Internet of Things
pH	Acidity/alkalinity indicator
EC	Electrical Conductivity (nutrient strength indicator)
TDS	Total Dissolved Solids

VOC	Volatile Organic Compounds
ESP32	Wi-Fi-enabled microcontroller
UI	User Interface
FR	Functional Requirement
NFR	Non-Functional Requirement
DD	Design Document
SRS	Software Requirements Specification

1.4 Overview

Organic NFT hydroponic systems require continuous monitoring due to variable nutrient characteristics of compost tea. At this stage, the project focuses on building the physical NFT system and establishing reliable sensor monitoring with cloud visualization, forming the foundation for later intelligent control.

2 Overall Descriptions

The Automated Organic NFT Hydroponic Monitoring System is designed as a modular embedded platform that supports real-time sensing, data acquisition, and remote monitoring for organic hydroponic cultivation using compost tea. At the current stage of development, the system primarily functions as a monitoring and data collection solution, establishing a stable foundation for future automation and control.

The system is cantered around an ESP32 microcontroller, which coordinates continuous data acquisition from multiple sensors including pH, EC, temperature, humidity, water level, gas, and color sensors. These sensors capture critical environmental and nutrient-related parameters required to understand the behaviour of organic nutrient solutions, which are inherently more variable than inorganic alternatives. The collected data is processed locally for validation and then transmitted to a cloud platform for visualization and storage.

Rather than aggressive automation at this stage, the design emphasizes reliability, safe operation, and data accuracy. The NFT structure, water circulation system, and grow channels have been physically implemented and tested to ensure consistent nutrient flow and stable operating conditions. Sensor readings are verified through serial monitoring and dashboard visualization to confirm correctness before advancing to closed-loop control in later phases.

In addition to local operation, the system supports remote visibility through cloud-based synchronization, allowing users to monitor real-time sensor readings via a mobile or web dashboard. Historical data logging enables trend observation and performance evaluation without direct physical access to the system. The overall architecture is intentionally modular and scalable, ensuring that advanced features such as compost maturity classification, fuzzy logic–based nutrient dosing, and automated pump control can be integrated in future stages without major redesign.

2.1 Product perspective

This component is designed as a smart monitoring and control add-on for an organic NFT hydroponic system used in capsicum chili cultivation. Existing IoT-based hydroponic solutions primarily focus on monitoring basic parameters such as pH, EC, and temperature, with simple remote visualization. However, they generally do not address the unique variability of organic nutrient solutions, particularly compost tea, which can change rapidly in quality and directly affect plant growth.

At the current development stage, this component functions as a real-time sensing and cloud-based monitoring system, providing continuous visibility of environmental and nutrient-related conditions. The physical NFT structure, water circulation, grow channels, and multi-sensor network have been implemented to establish a reliable operational foundation.

Novelty / creative gap (planned integration):

The long-term objective of this component is to introduce real-time compost tea maturity classification using sensor fusion of gas (VOC proxy) and color data, and to use this classification as an input for closed-loop fuzzy logic control. This approach aims to reduce manual judgement, handle organic variability more effectively, and improve nutrient stability in organic NFT systems. These intelligent control features are conceptually designed and will be implemented in later phases of the project.

2.1.1 System interfaces

- ESP32 microcontroller firmware handles sensor data acquisition and local processing
- Connectivity to a cloud database and dashboard via Wi-Fi
- System continues basic monitoring locally even if cloud connectivity is temporarily unavailable

2.1.2 User interfaces

- Cloud-based dashboard (web/mobile friendly)
- Live visualization of sensor readings (pH, EC, temperature, humidity, water level, gas, color)
- Basic status indicators for system monitoring
- Manual override interface for pumps and actuators (implemented for testing and monitoring phase)

2.1.3 Hardware interfaces

- **Sensors:** pH, EC, temperature & humidity, ultrasonic water-level sensor, gas sensor (VOC proxy), color sensor
- **Actuators:** circulation pump and relay-controlled pumps (manual control at current stage)
- **Hydroponic structure:** NFT grow channels, nutrient tank, continuous flow system

2.1.4 Software interfaces

- ESP32 firmware developed using Arduino IDE (C/C++)
- Firebase Realtime Database for data logging and dashboard updates
- Mobile or web-based dashboard for real-time visualization

2.1.5 Communication interfaces

- Wi-Fi communication between ESP32 and cloud database
- REST/HTTP-based data transmission for sensor uploads and control signals
- Lightweight data packets used to support frequent sensor updates

2.1.6 Memory constraints

The ESP32 has limited RAM and flash memory; therefore, sensor data structures and logs are kept compact. Future compost classification logic is planned to be lightweight (rule-based or small-scale model) to remain within hardware constraints.

2.1.7 Operations

Normal operations:

- Power-on and sensor warm-up
- Continuous sensing and data acquisition
- Upload of sensor readings to cloud database
- Real-time visualization on dashboard

Special operations:

- Sensor calibration mode
- Pump testing and maintenance mode
- Safe monitoring mode if a sensor provides invalid readings

2.1.8 Site adaptation requirements

- Suitable for small-scale greenhouses or protected farming environments in Sri Lanka
- Designed to tolerate weak or unstable internet connectivity by prioritizing local monitoring
- Compatible with low-power or solar-assisted energy setups with future optimization

2.2 Product functions

The major functions of the component at the current stage of development are as follows:

1. **Acquire multi-sensor data** related to the hydroponic environment and organic nutrient solution, including pH, EC, temperature, humidity, water level, gas, and color readings.
2. **Validate and preprocess sensor data** to ensure reliable monitoring and prevent invalid readings from affecting system operation.
3. **Upload real-time sensor data to a cloud database**, enabling remote access and historical data logging.
4. **Display live sensor readings on a cloud-based dashboard**, allowing users to monitor system conditions remotely.
5. **Support manual control of pumps and actuators** through the dashboard for testing and monitoring purposes.
6. **Provide a foundation for future intelligent features**, including compost tea maturity estimation and fuzzy logic–based nutrient and pH control, which are planned for later project phases.

2.3 User characteristics

Primary users:

- Smallholder farmers and greenhouse growers who require a simple, clear interface to monitor system conditions without technical complexity.

Secondary users:

- Technicians or system maintainers responsible for sensor installation, calibration, and troubleshooting.

Evaluators:

- Supervisors and evaluation panel members who require clear documentation of requirements, system design, and validation approach.

2.4 Constraints

The system design and implementation are subject to the following constraints:

- Sensor drift, particularly in pH, EC, and gas sensors, requiring periodic calibration.
- Variability in organic compost tea composition across different batches.
- Power limitations when operating under solar or battery-based systems.
- Unreliable or weak internet connectivity in rural farming environments.
- Risk of pump, tubing, and filter clogging due to organic nutrient solutions.
- Limited memory and processing capacity of the ESP32 microcontroller.

2.5 Assumptions and dependencies

The following assumptions are made during system design and implementation:

- Compost tea is prepared using a reasonably consistent method so that sensor trends are meaningful.
- Sensors are correctly installed, waterproofed, and maintained.
- Farmers or operators perform basic cleaning and calibration when required.
- Internet connectivity may be intermittent; therefore, basic local monitoring must continue even when cloud access is unavailable.

2.6 Apportioning of requirements

The requirements are implemented in a phased priority order aligned with project milestones:

1. **Core sensing and stable data acquisition** (completed at Progress Presentation 1).
2. **Compost maturity identification** using gas and color sensor trends (planned).
3. **Fuzzy logic–based control** for nutrient dosing, irrigation, and pH balancing (planned).
4. **Cloud logging and dashboard visualization** (implemented at Progress Presentation 1).
5. **Alerts, automation refinement, and reliability improvements** (planned for later stages)

3 Specific requirements

3.1 External Interface Requirements

3.1.1 User Interfaces

The embedded system provides two primary user interfaces:

1. Cloud-Based Dashboard Interface.

   The system connects to a cloud-based dashboard that displays real-time sensor readings collected from the organic NFT hydroponic setup. The dashboard presents values such as pH, EC, temperature, humidity, water level, gas sensor readings, and color sensor readings in a clear and user-friendly format. This interface is designed to support non-technical users, such as smallholder farmers, by providing simple visual monitoring of system conditions.

2. Local LCD Interface

   A 16×2 I2C LCD is integrated with the ESP32 to provide on-site monitoring. The LCD cycles through key sensor readings and system status messages, allowing users to verify system operation without accessing the cloud dashboard. This interface is particularly useful during installation, testing, and situations where internet connectivity is unavailable.

Future user interface enhancements, such as compost maturity indicators, alerts, and intelligent recommendations, are planned for later project stages.

3.1.2 Hardware Interfaces

The embedded system interfaces with the following hardware components:

- ESP32 microcontroller, which acts as the central control and communication unit.
- Environmental sensors, including a temperature and humidity sensor to monitor greenhouse conditions.
- Nutrient monitoring sensors, including pH and EC sensors for organic nutrient solution assessment.
- Ultrasonic sensor, used to estimate water or nutrient tank levels.

- Gas sensor, used to measure gas concentration trends associated with compost tea activity.
- Color sensor, used to capture color characteristics of compost tea.
- Relay modules, used to manually control pumps during the monitoring phase.
- NFT hydroponic hardware, including grow channels, nutrient tank, circulation pump, tubing, and filtration components.
- All sensors and actuators interface with the ESP32 using standard embedded communication methods such as GPIO, I2C, UART (RS485), and analog input.

3.1.3 Software Interfaces

The embedded system interacts with the following software components:

- ESP32 firmware, developed using the Arduino IDE and written in C/C++, responsible for sensor data acquisition, validation, and communication.
- Firebase Realtime Database, which serves as the cloud backend for data storage, synchronization, and dashboard access.
- Dashboard application, which retrieves sensor data from Firebase and displays it to users in real time.
- Development and debugging tools, including serial monitoring tools used for testing and validation during development.

At the current project stage, software interfaces support monitoring and manual interaction. Intelligent control and classification software modules are planned for future integration.

3.1.4 Communication Interfaces

The embedded system uses Wi-Fi communication as the primary communication interface between the ESP32 and the cloud platform.

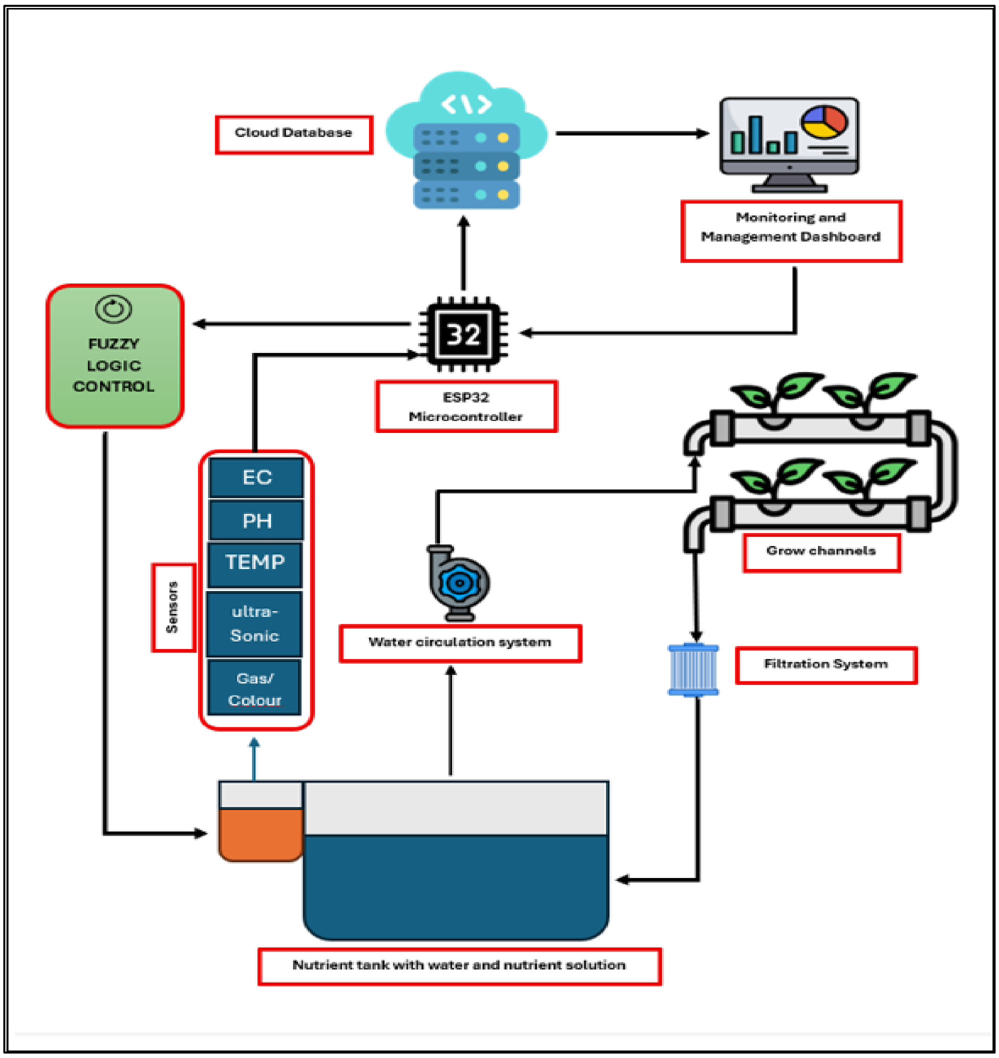
Key communication characteristics include:

- HTTP-based data transmission for uploading sensor readings to the cloud database.
- Periodic data updates to support near real-time monitoring.
- Two-way communication, allowing the system to receive basic control commands (e.g., pump ON/OFF) from the dashboard during testing.

The system is designed to handle intermittent connectivity. If the internet connection is unavailable, local monitoring continues, and data synchronization resumes automatically once connectivity is restored. Support for optimized or alternative communication methods may be introduced in later stages if required.

3.2 Architectural Design

3.2.1 High level Architectural Design



3.2.2 Hardware and software requirements with justification

Hardware Requirements

The hardware platform must support reliable sensing and stable operation in an organic NFT hydroponic environment. At the current stage, the following hardware capabilities are required and partially implemented:

- **Reliable sensing for pH, EC, and environmental parameters** to continuously monitor nutrient strength and growing conditions, which is critical due to the variability of organic compost tea.
- **Gas and color sensing capability** to capture indicators related to compost tea activity and appearance. These sensors are implemented at the data acquisition level and provide the foundation for future compost maturity estimation.
- **Circulation pumps and relay-controlled actuators** to maintain continuous nutrient flow through NFT channels and to allow manual control during monitoring and testing.
- **Aeration support through system design**, including sufficient nutrient flow and air exposure, to improve root oxygenation and avoid anaerobic stress in capsicum plants.

These hardware choices are justified by their low cost, availability, and suitability for small-scale greenhouse deployment in Sri Lankan conditions.

Software Requirements

The software system must support reliable data acquisition, processing, and visualization while remaining lightweight and extensible.

At the current implementation stage, the software supports:

- **Embedded firmware on the ESP32** for reading sensor data, validating values, and handling communication.
- **Cloud-based data logging** to store sensor readings for monitoring and analysis.
- **Dashboard connectivity** to visualize real-time system conditions.

Planned software extensions include:

- **Compost maturity classification logic**, using gas and color sensor data in a lightweight rule-based or small-model approach suitable for ESP32 memory constraints.
- **Fuzzy logic decision support**, to enable automated nutrient dosing, irrigation scheduling, and pH balancing in later stages.

3.2.3 Risk Mitigation Plan with alternative solution identification

Several technical and operational risks have been identified, along with corresponding mitigation strategies:

- **Sensor drift (pH, EC, gas sensors):**
  Mitigated through scheduled calibration, value range checks, and use of safe default values when abnormal readings are detected.

- **Noise in gas and color sensor readings:**
  Mitigated by applying averaging, smoothing filters, and consistent sampling timing during data collection.

- **Clogging due to organic nutrient solutions:**
  Mitigated by in-line filtration, periodic flushing of pipes, and limiting continuous pump operation.

- **Network connectivity loss:**
  Mitigated by allowing local sensing and monitoring to continue without cloud access, with data synchronization resuming once connectivity is restored.

- **Pump or actuator failure:**
  Mitigated by relay safety defaults, time-based limits on pump operation, and dashboard alerts when abnormal behaviour is detected.

- **Incorrect compost maturity interpretation (future stage):**
  Mitigated by cross-checking gas and color trends with supporting parameters such as EC and pH, and by allowing manual farmer override.

3.2.4 Cost Benefit Analysis for the proposed solution

The proposed component is designed with a strong focus on affordability and practicality for smallholder farmers.

From a cost perspective:

- The system uses low-cost sensors and an ESP32 microcontroller, reducing initial hardware investment.

- Organic compost tea is locally available and reduces dependency on expensive chemical nutrient solutions.

From a benefit perspective:

- Continuous monitoring reduces the need for frequent manual testing and observation.
- Improved visibility of nutrient and environmental conditions supports more stable capsicum growth.
- The modular design allows the system to scale to additional NFT channels or future automation without major redesign.

  Overall, the expected benefits in terms of reduced labour, lower input costs, and improved yield stability justify the implementation cost and support the feasibility of wider adoption.

3.3 Performance requirements

The system must meet specific performance requirements to ensure reliable monitoring and to support future intelligent control in an organic NFT hydroponic environment.

At the current stage, the system is required to:

- Provide near real-time sensor updates, allowing early detection of changes in environmental and nutrient conditions. Sensor readings should refresh frequently enough to support effective monitoring and timely user response.
- Ensure stable and consistent data acquisition, avoiding missing or corrupted readings during continuous operation.
- Support timely dashboard updates, so that users can observe system behaviour remotely and apply manual overrides when required.

For later stages of the project, additional performance targets are defined:

- **Compost tea maturity classification** should achieve high practical accuracy during experimental trials, with a target accuracy of approximately 90% under controlled conditions.

- **Automated control actions** must avoid rapid oscillations or unstable behaviour by applying safe timers, gradual adjustments, and protective rule limits during nutrient dosing and pH balancing.
- **Alert and notification mechanisms** must respond quickly to abnormal conditions to minimize risk to plants and equipment.

3.4 Design constraints

The design and implementation of the system are subject to several technical and environmental constraints.

- The system must operate within the memory and processing limitations of the ESP32, requiring lightweight firmware design and efficient data handling.
- The system must be suitable for low-power operation, particularly when deployed with solar or battery-based power sources, by optimizing sensor sampling and communication frequency.
- The system must tolerate weak or intermittent internet connectivity, which is common in rural agricultural environments, by prioritizing local monitoring and resuming cloud synchronization when connectivity is restored.
- Any chemical inputs used for pH correction must be safe, minimal, and environmentally responsible, in line with the sustainability goals of organic hydroponic farming.

These constraints guide both the current monitoring implementation and the design of future automation features, ensuring that the system remains practical, safe, and deployable in real-world farming conditions.

3.5 Software system attributes

3.5.1 Reliability

The software system is designed to operate reliably in a real-world organic NFT hydroponic environment. At the current stage, the system ensures that local sensing and monitoring continue even during temporary cloud or network failures. Sensor readings are handled safely to avoid system crashes due to invalid or missing data.

To protect hardware and plants, safe actuator limits and default states are enforced for pumps and relays, ensuring that uncontrolled operation does not occur during unexpected conditions. The system architecture supports the future integration of watchdog timers and additional fault-handling mechanisms to further enhance reliability.

3.5.2 Availability

The system aims to maintain high availability during plant growth cycles, as continuous monitoring is essential in hydroponic cultivation. The ESP32 firmware is designed to start quickly after power interruptions and resume sensor data acquisition and cloud synchronization with minimal delay.

Short restarts or connectivity interruptions do not require manual reconfiguration, allowing the system to recover automatically and continue normal operation, which is important for long-term greenhouse deployments.

3.5.3 Security

Basic security measures are incorporated to protect system data and prevent unauthorized control actions. Cloud database access is managed through authentication and database access rules, ensuring that only authorized devices and users can read sensor data or issue control commands.

Dashboard access is restricted to permitted users, reducing the risk of accidental or malicious pump activation. While advanced security mechanisms are beyond the scope of the current implementation, the system design allows for stronger access control and encryption to be added in later stages.

3.5.4 Maintainability

The software is developed with maintainability and extensibility in mind. The firmware is organized into modular components, separating sensor data acquisition, data validation, cloud communication, and control logic. This structure simplifies debugging, testing, and future feature integration.

Sensors are selected and interfaced in a way that allows easy replacement and recalibration without major changes to the firmware. Calibration and maintenance steps are documented to support long-term operation and to reduce dependency on specialized technical knowledge.

3.6 Other requirements

In addition to functional and performance-related requirements, the system must satisfy several supporting requirements to ensure usability, safety, and long-term research value.

The system must support continuous data logging of sensor readings in a structured cloud database. This logged data should enable growth trend analysis, comparison across growth stages, and evaluation of system performance over time through historical graphs and summaries.

From an ethical perspective, the project involves plant-based experimentation only. No human or animal subjects are used at any stage of development or testing. As such, the system does not raise ethical concerns related to human or animal safety and complies with standard academic research guidelines.

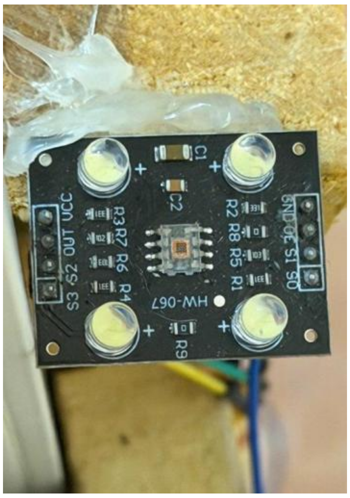
From an environmental perspective, the system is designed to support sustainable and eco-friendly farming practices. By enabling organic nutrient use, continuous monitoring, and reduced reliance on chemical inputs, the system contributes to lower chemical usage, improved resource efficiency, and environmentally responsible hydroponic cultivation.

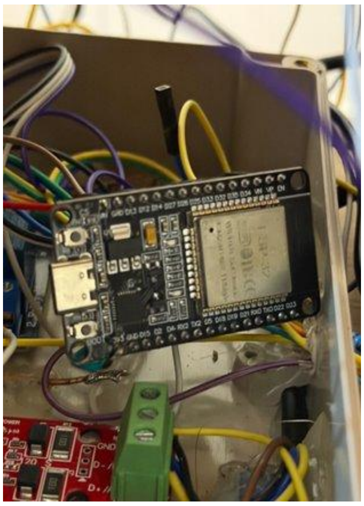
4 Supporting information

4.1 Appendices

4.1.1 Organic NFT Hydroponic System Prototype Setup and both Organic and Inorganic NFT Hydroponic Systems

4.1.2 Hardware Components and Specifications:





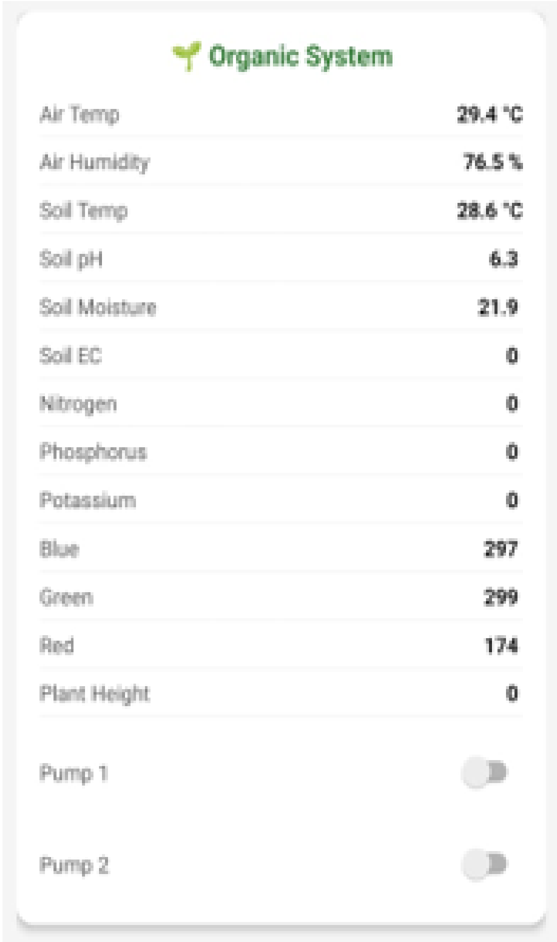
4.1.3 Firmware Code Snapshot.

```
String firebaseHost = "https://hydroponic-3ddcf-default-rtdb.firebaseio.com";
String apiKey = "CB4lzSLIQEiWluk4gPxjY0YL1MDA14xWfJgvviVt";
```

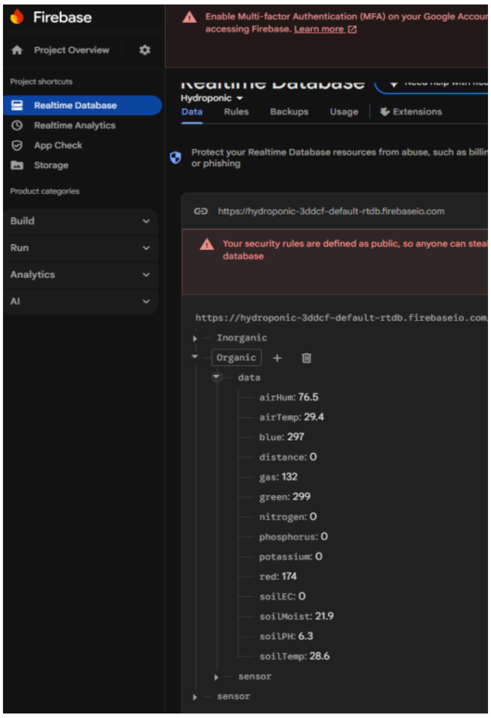
```
// ================= LCD ================
LiquidCrystal_I2C lcd(0x27, 16, 2);

// ================= RS485 NPK ================
#define RXD2 16
#define TXD2 17

uint8_t byteRequest[8] = {0x01, 0x03, 0x00, 0x00, 0x00, 0x07, 0x04, 0x08};
uint8_t byteResponse[32];

float soilMoist=0, soilTemp=0, soilPH=0;
int soilEC=0, nitrogen=0, phosphorus=0, potassium=0;

// ================= TCS230 ================
#define S0 14
#define S1 12
#define S2 27
#define S3 26
#define OUT_PIN 25

// ================= DHT22 ================
#define DHTPIN 4
#define DHTTYPE DHT22
DHT dht(DHTPIN, DHTTYPE);

// ================= ULTRASONIC ================
#define TRIG_PIN 18
#define ECHO_PIN 19

// ================= MQ GAS ================
#define MQ_PIN 34

// ================= RELAYS ================
#define RELAY_PUMP1 32
#define RELAY_PUMP2 33
```

```
// ================= SETUP ================
void setup() {
  Serial.begin(115200);
  Serial2.begin(4800, SERIAL_8N1, RXD2, TXD2);
  delay(2000);
  Serial.println("Soil NPK Sensor - Stable Reading");

  Wire.begin(21,22);

  // LCD init
  lcd.begin(16,2);
  lcd.backlight();
  lcd.print("System Starting");
  delay(1000);

  Serial.println("Soil NPK Sensor - Stable Reading");
  lcd.clear();
  lcd.print("Soil NPK Sensor - Stable Reading");

  pinMode(S0,OUTPUT); pinMode(S1,OUTPUT);
  pinMode(S2,OUTPUT); pinMode(S3,OUTPUT);
  pinMode(OUT_PIN,INPUT);
  digitalWrite(S0,HIGH); digitalWrite(S1,LOW);

  pinMode(TRIG_PIN,OUTPUT);
  pinMode(ECHO_PIN,INPUT);

  pinMode(RELAY_PUMP1,OUTPUT);
  pinMode(RELAY_PUMP2,OUTPUT);
  digitalWrite(RELAY_PUMP1,HIGH);
  digitalWrite(RELAY_PUMP2,HIGH);

  dht.begin();

  WiFi.begin(WIFI_SSID,WIFI_PASSWORD);
  while(WiFi.status()!=WL_CONNECTED){delay(1000);}
  lcd.clear();
  lcd.print("WIFI Connected!");
  updatePumpStatus();
}
```

4.1.4 Cloud Dashboard Screenshots.



4.1.5 Test Results (Firebase data logs)



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